Software Requirements Specification

For

Student Feedback Review System

Version 1.0

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1. Introduction

1.1. Purpose

This document defines the functional and non-functional requirements for the Student Feedback Review System (SFRS) Version 1.0 for EduTech Institute Singapore (ETIS). It serves as a reference for development and a basis for verification and validation. Project management, cost, and implementation details are excluded.

Please refer to Appendix D: Declarations for AI and Plagiarism Declarations.

1.2. Document Conventions

This SRS follows the structure and recommended practices outlined in IEEE Std 830-1998 and adopts a **template style derived from Karl E. Wiegers' SRS template**.

The following conventions apply throughout this document:

- The following imperative verbs are used consistently: *Shall* (mandatory requirement), *Should* (desirable but non-mandatory requirement), and *May* (optional capability).
- Requirement categories are identified using the following prefixes: UR-TYPE-XX (User Interface Requirements), ITR-TYPE-XX (Technical Requirements), FR-ROLE-XX (Functional Requirements), NFR-TYPE-XX (Non-Functional Requirements), BR-XX (Business Rules), OR-TYPE-XX (Other Requirements).
- Requirements are mapped in the Initial Requirements Traceability Matrix (iRTM) to ensure traceability. **However, additional technical requirements derived from industry best practices and implementation considerations may not appear in the iRTM**, as they represent supporting design or constraints.

1.3. Intended Audience and Reading Suggestions

This document is intended for stakeholders involved in the specification, development, verification, governance, and oversight of the SFRS. Reading recommendations are as follows:

- **Executive and governance stakeholders** should begin with Sections 1 and 2 to understand document scope and system context before reviewing high-level system features in Section 4.
- **Developers and architects** should review Sections 2 and 3 before proceeding to detailed System Features in Section 4.
- **Testers** should focus primarily on Sections 3, 4, and 5, where verifiable requirements are specified.
- **Compliance reviewers** should examine Sections 2.5, 5.3, and 5.5 for constraints, security requirements, and business rules.

1.4. Product Scope

The SFRS is an institutional academic feedback management system designed to digitize and automate the collection, processing, and analysis of student feedback at ETIS.

The primary objectives of the system, as defined in ETIS's Project Specification (PS), are to:

- Goal 1: Increase student feedback participation rates
- Goal 2: Reduce feedback processing turnaround time
- Goal 3: Provide aggregated and analyzable feedback insights
- Goal 4: Support scalable digital operations aligned with projected institutional growth.

1.5. References

The following documents and standards are referenced within this SRS:

1. EduTech Institute Singapore. *Project Specification: Student Feedback Review System*, Version 1.0, 30 January 2026. Link: https://tinyurl.com/mr7duj7z
2. NovaStack Pte Ltd Singapore. *Stock Template: Student Feedback Review System*, Version 1.0, 30 January 2026. Link: https://tinyurl.com/yc38hcbt
3. Wiegers, Karl E. *Software Requirements*, Microsoft Press, 1999. Link: https://tinyurl.com/bahuy3m3
4. Personal Data Protection Commission (Singapore). *Personal Data Protection Act (PDPA)*, current revision. Link: https://sso.agc.gov.sg/Act/PDPA2012
5. ETIS Institutional Data Governance and Retention Policies (Internal Reference).

2. Overall Description

2.1. Product Perspective

The SFRS is a new institutional academic feedback management system commissioned by ETIS. The system is to act as a replacement for the existing manual, paper-based student feedback collection process with a centralized digital platform.

The SFRS operates as a standalone application within ETIS's institutional IT ecosystem but interfaces with existing institutional systems for data validation and user identity verification. Specifically, the system interacts with See Fig 1: Context diagram (for visualization):

- The Student Registrar System (for enrolment validation);
- The Human Resources (HR) System (for faculty status verification);
- Institutional authentication services (for user identity and access control).

2.2. Product Functions

At a high level, the SFRS shall provide the following major functional capabilities. Please refer to Section 4: System Features for detailed explanations.

1. Role-Based Access Control (Goal 4) - Restricts system actions and data access based on predefined user roles.
2. User Account Access (Goal 4) - Allows authorized users to securely log in and manage their system accounts.
3. Administrative Audit Logging (Goal 4) - Records administrative activities for accountability and security review.
4. Feedback Period Management (Goal 1, 2) - Enables administrators to create and control when feedback collection periods are open.
5. Feedback Question Management (Goal 1) - Allows administrators to create, update, or remove feedback questions used in surveys.
6. Feedback Submission (Goal 1) - Allows students to submit feedback responses during active feedback periods.
7. Feedback Review Access (Goal 3) - Provides authorized staff with access to review collected feedback results.

2.3. User Classes and Characteristics

Academic Administrators (Primary User Class)
Priority: Highest (Operational Core), moderate to high usage during academic term setup, in charge of full configuration and management privileges, system-wide administrative control.

Faculty Members (Primary User Class)
Priority: High (Core System Users), moderate usage after feedback periods, view-only access to aggregated reports for assigned courses.

Students (Primary User Class)
Priority: High (Core System Users), high concurrent usage during feedback periods, submit feedback for analysis.

Refer to Figure 2: High Level Data Flow to see how each user class interacts with the Product Functions

2.4. Operating Environment

The SFRS shall operate in a web and mobile compatible environment **using the latest, actively supported technology stacks as of 6 March 2026**:

- **Web Application:** Python with Django 5.x (latest major supported series)
- **Database:** MySQL 8.1+
- **Frontend:** HTML5, CSS3, JavaScript (ES2023/ESNext)
- **Cloud Infrastructure:** AWS (latest stable services in Singapore region)
- **Browser Support:** Latest two stable versions of Chrome, Edge, Safari, and Firefox
- **Operating Systems:** Modern desktop and mobile OS families (Windows 11+, macOS Sonoma+, iOS 17+, Android 14+)
- **Network Security:** HTTPS using TLS 1.3

The system shall coexist with institutional IT infrastructure and shall not interfere with the operation of registrar, HR, or authentication systems.

2.5. Design and Implementation Constraints

The following constraints limit available design and implementation options.

1) **Regulatory and Compliance Constraints**
   - Compliance with Singapore's Personal Data Protection Act (PDPA).
   - Adherence to ETIS institutional data governance and retention policies.
   - Support for auditability and accountability of administrative actions.
2) **Integration Constraints**
   - Mandatory integration with the existing Student Registrar System and Human Resource System.
   - Use of institutional authentication mechanisms for access control.
   - Use of standardized communication protocols for external system interactions.
3) **Maintainability and Support Constraints**
   - The system shall be designed to support future maintenance and enhancement activities.
   - Adequate documentation shall be provided to enable institutional technical staff to support operational continuity.
   - Long-term maintenance and cost arrangements are outside the scope of this SRS.

2.6. Assumptions and Dependencies

The following assumptions underlie the requirements specified in this document:

- Institutional registrar and HR systems provide accurate and up-to-date data.
- Institutional authentication services remain operational and accessible.

- Users possess basic familiarity with web-based systems.
- Network connectivity is available during defined feedback periods.
- Institutional policies governing feedback processes remain stable during development.
- Feedback collection is conducted on a per-term basis and applied uniformly across all subjects and classes

The system is dependent on:

- Availability and accessibility of external institutional data sources.
- Continued institutional compliance requirements under PDPA.
- Approval and support from academic governance bodies.

2.7. Change Mitigation Strategy

This section defines the structured mechanisms by which requirement changes shall be anticipated, evaluated, controlled, and incorporated during the Software Development Life Cycle (SDLC).

2.7.1. Anticipatory Contingency Requirements

Anticipatory contingency requirements are pre-agreed conditional requirements documented within the SRS but activated only when specific triggering conditions occur.
These contingencies are defined in Section 6: Change Mitigation Controls and shall:

- Be clearly identified with unique requirement IDs.
- Specify activation conditions.w
- Define impact scope (functional, non-functional, data schema, integration, etc.).

2.7.2. Open Issues

Not applicable, all requirements have been finalised in the Analysing Phase in the Analysing Report.

2.7.3. High-Risk Change Control

High-risk change control applies to modifications that may:

- Significantly alter system architecture,
- Impact contractual cost or schedule,
- Affect core system goals,
- Introduce regulatory or compliance risk.

As well as modifications to requirements that have been designated as high risk (risk score > 15 in risk_ analysis/risk_analysis_matrix_table).

All high-risk changes shall undergo the following process:

1. Formal Change Request Submission.
2. Impact Analysis (technical, financial, schedule, quality).
3. Risk Assessment.
4. Approval by designated authority (Project Manager and Customer Representative).
5. Update of SRS, iRTM, Project Plan and Version history log:

Changes deemed to constitute entirely new requirements (i.e., not derivable from existing SRS content) may be rejected or deferred to future project phases. Refer to the Change Control Agreement in the repository for more details or Appendix E: Change Control Agreement.

3. External Interface Requirements

These requirements assume the use of the latest stable and actively supported technologies as of 6 March 2026.
The link to the full **iRTM spreadsheet** can also be found here.

3.1. User Interfaces

The SFRS provides a responsive user interface designed to maintain a uniform user experience across multiple hardware platforms, including windows based browsers and mobile applications for Android and iOS. The interface would utilise a role based dashboard system logic where layout and available functions would dynamically adjust based on the authenticated use. The sidebar content would be restricted by roles where different users will have different sidebar functionality. The sample screen for login page, feedback form page and analytics images are shown in Appendix C.

3.1.1 Logical Characteristics

Standard buttons and functions
Some functions and buttons that appear on the navigation bar for every screen except for the login page will include a global help button to provide a link to help guide the users and user profile button that links to user's profile details.

Keyboard shortcuts
To support efficiency, the desktop interface supports tab, enter and esc. Tab is used to cycle through the form fields or navigation buttons. Enter is used to submit the current form page or to confirm actions. Finally, Esc is used to close active modals or help overlays.

.**Error message and validation standards**
The validation standards of the system are:

- **Inline mandatory validation**: System will block submission and would display a red highlight on mandatory fields or questions if they are unanswered.

- **Policy violation alerts**: When a user submits credentials that fail institutional policy such as if password complexity is too low, the system would return a specific error message describing the exact policy violation.

General interface requirements
- **UR-GUI-01**: The system shall support latest and prior versions of chrome, firefox, safari and edge on desktop and mobile platform

- **UR-GUI-02:** The interface shall be fully responsive maintaining usability on screen widths ranging from 360px on mobile to 1920px on desktop.

- **UR-GUI-03:** The system screens other than the login page shall have a navigation bar that contains the ETIS logo, page title and global icons for user profile and help.

- **UR-GUI-04:** The system shall have a sidebar that will collapse to a hamburger for screens smaller than 448px

- **UR-GUI-05:** The system shall use primarily ETIS Blue (#004A99) for headers and primary action buttons, neutral slate gray (#F8FAFC) for background areas to reduce visual fatigue.

- **UR-GUI-06:** The system shall use "Inter" or "Open Sans" sans serif font families standardised for all text

Student interface requirements

- **UR-STU-01**: The system interface shall provide a text based feedback field with minimum 50 words and maximum 500 words for sentiment analysis
- **UR-STU-02:** The system interface shall have a rating field with course ratings of 1 to 5.

Faculty interface requirements

- **UR-FAC-01**: The faculty dashboard shall display aggregated summary cards for Average Rating and Total Submissions.
- **UR-FAC-02:** The system shall visualise the rating distributions using bar charts and qualitative sentiment using word clouds

Academic administrators requirements

- **UR-ADM-01**: The system dashboard shall display system-wide metrics, including overall submission rates and system health status.
- **UR-ADM-02:** The system shall display line graphs that track submission trends over time.

3.2. Hardware Interfaces

The SFRS provides a responsive experience across multiple hardware environments by decoupling the client-side interface from the backend cloud infrastructure.

3.2.1 Supported Device Types

The fully functional responsive web interface will be compatible with standard PC hardware running windows based browsers and dedicated hardware optimised interfaces are also provided for mobile devices operating on both Android and iOS platforms.

3.2.2 Nature of Data and Control Interactions

Depending on what the user role is, the interface and data will dynamically change to the user's role. The software interfaces shall also support standard desktop input and mobile specific interactions to facilitate data entry and submission of feedback.

- **Hardware Technical Requirements:**
  - **IT-SA-01:** The system shall provide a responsive web interface compatible with Windows-based browsers.
  - **IT-SA-02:** The system shall provide dedicated mobile application interfaces for both Android and iOS platforms.
  - **IT-SA-03:** The system shall enforce uniform user access roles across all supported hardware platforms.

3.3. Software Interfaces

The system establishes secure boundaries for data exchange with institutional systems to ensure data integrity and automated synchronisation.

3.3.1 Institutional Database Integration

The system establishes an authorised interface to synchronise records from external sources such as student Registrar system where enrollment data and subject class pairings are retrieved and Human Resources system where faculty status and faculty subject assignments are retrieved.

3.3.2 Synchronisation Governance

The system defines specific synchronisation boundaries that prevents unauthorised data overwrites from external sources and every data synchronisation event will be logged to maintain data authority and interface ownership.

- **Software Technical Requirements:**
  - **IT-SI-01:** The system shall establish an authorised interface to synchronise data with the institutional Registrar and HR databases.
  - **IT-SI-02:** The system shall define specific synchronisation boundaries to prevent unauthorised data overwrites from external sources.
  - **IT-SI-03:** The system shall log all data synchronisation events to verify interface ownership and data authority.

3.4. Communications Interfaces

3.4.1 Communication standards

The system maintains secure communication using protocols such as HTTP with TLS 1.3 ensuring confidentiality and integrity of data between the client and the server. HTTP protocol is used to facilitate all data synchronisation tasks with external systems such as the ETIS student Registrar and HR systems. The system also will implement an automated session management logic which includes 30 second timeout thresholds and maximum of 3 retransmission attempts therefore maintaining stable connections with the external systems and databases during periods of high network instability.

- **Communication Technical Requirements:**
  - **IT-CI-01:** The system shall encrypt all external communications using HTTPS with TLS 1.3
  - **IT-CI-02:** The system shall utilise the HTTP protocol to facilitate all data synchronisation tasks with the ETIS Student Registrar and HR systems
  - **IT-CI-03:** The system shall perform a maximum of 3 automated retransmission attempts when a communication timeout occurs.
  - **IT-CI-04:** The system shall implement a mandatory 30 second timeout for all external communication requests.

4. System Features

This section organises all functional requirements (FRs) into seven discrete capabilities that directly implement ETIS's Business Operation (BOR) requirements, presenting each feature as a cohesive service with its priority, user-system response sequences, and verifiable "shall" statements. The Inceptive Requirements Traceability Matrix (iRTM) provides bidirectional traceability from each PS BOR to its expanded SRS FRs demonstrating 1:N cardinality while preserving Goal, Purpose and Value (GPV) intent and stakeholder ownership, while ensuring the vendor's interpretation aligns with customer expectations. Additionally, see Figure 3: Use Case Diagram in the Appendix structurally validates this organisation by mapping each system feature to an actor-initiated use case package and see Annex A Data Dictionary for expected fields declared by ETIS in the Project Specification.

A link to the **iRTM spreadsheet** can be found here.

4.1. Role-Based Access Control

4.1.1. Description and Priority

This feature governs how the system assigns and enforces access privileges across all user types. Academic Administrators assign each user a single role upon account provisioning, and the system restricts all functions and information access based on that role at all times. See Table 1: Role-Based Access Control Response Sequence in Appendix B for the system response sequence.

UC1: Role-Based Access Control
Priority: High (Essential)

4.1.2. Functional Requirements

- **FR-AC-01**: The system shall allow Academic Administrators to assign each user exactly one role (Student, Faculty, or Administrator) upon account provisioning.
- **FR-AC-02**: The system shall restrict system functions and information access based on the user's assigned role.
- **FR-AC-03**: The system shall allow Academic Administrators to create, update, and deactivate user accounts.

4.2. User Account Access

4.2.1. Description and Priority

This feature enables authorised users to manage their own authentication credentials without requiring administrative intervention. Credential changes are subject to identity verification and institutional password policies. Users have no self-service access to their own roles or access privileges. See Table 2: User Account Access Response Sequence in Appendix B for the system response sequence. Additionally see Annex A Data Dictionary for expected credential fields.

UC2: User Account Access
Priority: High (Essential)

4.2.2. Functional Requirements

- **FR-AA-01**: The system shall allow authorised users to update their own password.
- **FR-AA-02**: The system shall require identity verification before applying password changes.
- **FR-AA-03**: The system shall prevent users from changing their own roles or access privileges.
- **FR-AA-04**: The system shall allow users to login with their valid school credentials including the school email and password.

4.3. Administrative Audit Logging

4.3.1. Description and Priority

This feature provides Academic Administrators with a traceable, tamper-protected record of all administrative actions performed on user accounts and access controls. Audit logs exist to support institutional governance, accountability, and regulatory compliance. See Table 3: Administrative Audit Logging Response Sequence in Appendix B for the system response sequence.

UC3: Administrative Audit Logging
Priority: High (Essential)

4.3.2. Functional Requirements

- **FR-AL-01**: The system shall record administrative actions performed on user accounts and access controls.

- **FR-AL-02**: The system shall include in each audit record the administrator identity, the timestamp, and the action performed.
- **FR-AL-03**: The system shall restrict audit log access to Academic Administrators only.

4.4. Feedback Period Management

4.4.1. Description and Priority

This feature enables Academic Administrators to create and govern the feedback collection window for each academic term. Feedback periods have defined start and end dates, apply uniformly across all subjects and classes in a term, and are subject to a single-active-period constraint. When a period closes, submissions are permanently disabled. See Table 4: Feedback Period Management Response Sequence in Appendix B for the system response sequence.

UC4: Feedback Period Management
Priority: High (Essential)

4.4.2. Functional Requirements

- **FR-FP-01:** The system shall allow Academic Administrators to create a feedback period by specifying a start date and end date for an academic term.
- **FR-FP-02**: The system shall enforce that at most one feedback period is active per academic term at any given time.
- **FR-FP-03**: The system shall apply the active feedback period uniformly across all subjects and classes within the academic term.
- **FR-FP-04**: The system shall permanently disable feedback submissions for a feedback period upon reaching its defined end date.
- **FR-FP-05**: The system shall automatically generate aggregated feedback analysis reports only for feedback periods that have been closed.
- **FR-FP-06:** The system shall allow Academic Administrators to update the start date, end date, or configuration of a feedback period before the feedback period start date has been reached.
- **FR-FP-07:** The system shall allow Academic Administrators to delete a feedback period before the feedback period start date has been reached.

4.5. Feedback Question Management

4.5.1. Description and Priority

This feature allows Academic Administrators to define the questions used during a feedback period, supporting both qualitative (open-ended) and quantitative (rating-scale) types. Once a feedback period becomes active, all associated questions are locked to ensure data consistency across all submissions. See Table 5: Feedback Question Management Response Sequence in Appendix B for the system response sequence. Additionally see Annex A Data Dictionary for expected fields for each feedback question.

UC5: Feedback Question Management
Priority: High (Essential)

4.5.2. Functional Requirements

- **FR-FQ-01**: The system shall allow Academic Administrators to define feedback questions of qualitative (open-ended text) and quantitative (rating-scale) types prior to the activation of a feedback period.
- **FR-FQ-02**: The system shall prevent modification or deletion of feedback questions once the associated feedback period has been activated.

- **FR-FQ-03**: The system shall allow Academic Administrators to designate individual feedback questions as mandatory or optional.
- **FR-FQ-04:** The system shall allow Academic Administrators to update feedback questions, including the question text, question type, and mandatory or optional designation, prior to the activation of the associated feedback period.
- **FR-FQ-05:** The system shall allow Academic Administrators to delete feedback questions prior to the activation of the associated feedback period.

4.6. Feedback Submission by Students

4.6.1. Description and Priority

This feature enables Students to submit feedback for Faculty members based on their enrolled subject-class pairings. Submissions are constrained to active feedback periods, assigned enrolments, one submission per pairing per period, and validated content. See Table 6: Feedback Submission by Students Response Sequence in Appendix B for the system response sequence. Additionally see Annex A Data Dictionary for expected feedback submission fields for the feedback question.

UC6: Feedback Submission by Students
Priority: High (Essential)

4.6.2. Functional Requirements

- **FR-FS-01**: The system shall allow Students to submit feedback only for faculty members associated with the subject-class pairings in which they are enrolled.
- **FR-FS-02**: The system shall restrict each Student to one feedback submission per faculty subject-class pairing per feedback period.
- **FR-FS-03**: The system shall associate each feedback submission with the Faculty member assigned to the selected subject-class pairing.
- **FR-FS-04**: The system shall permit feedback submission only during an active feedback period.
- **FR-FS-05**: The system shall reject feedback submissions where one or more mandatory questions have not been answered.
- **FR-FS-06**: The system shall validate submission content and reject submissions that fail institutional content validation rules.

4.7. Faculty Feedback Access

4.7.1. Description and Priority

This feature allows Faculty members to view aggregated feedback results and an analysed summary for their assigned subject-class pairings after the feedback period has closed. Student identities are not disclosed at any point, and Faculty have no ability to alter submitted feedback records. See Table 7: Faculty Feedback Access Response Sequence in Appendix B for the system response sequence. Additionally see Annex A Data Dictionary for expected fields for the feedback review summary.

UC7: Faculty Feedback Access
Priority: High (Essential)

4.7.2. Functional Requirements

- **FR-FA-01**: The system shall allow Faculty to access aggregated feedback results only for subject-class pairings to which they are assigned.
- **FR-FA-02**: The system shall make feedback results accessible to Faculty only after the associated feedback period has closed.
- **FR-FA-03**: The system shall present feedback results to Faculty in aggregated form and shall not disclose individual student identities.
- **FR-FA-04**: The system shall prevent Faculty from modifying or deleting any submitted feedback record.

5. Other Nonfunctional Requirements

5.1. Performance Requirements

The SFRS shall meet the following performance requirements to ensure responsiveness, scalability, and timely analytics processing during both normal operations and peak academic feedback periods. These requirements are defined to support high student participation rates and timely availability of feedback insights.

- **NFR-PR-01**: The system shall maintain a response time of less than 2 seconds for at least 95% of user-initiated transactions during peak feedback submission periods.
    - This requirement applies to:
        - User login
        - Loading feedback forms
        - Submitting feedback
        - Viewing dashboards

    - **Rationale:**
      During end-of-semester feedback windows, large numbers of students will access the system simultaneously. Maintaining sub-2-second response times prevents user frustration, reduces abandonment rates, and supports the institutional goal of increasing participation rates.

- **NFR-PR-02**: The system shall support a minimum of 10,000 concurrent student users during peak feedback periods without violating the response time requirement defined in NFR-PR-01.
    - **Rationale:**
      Feedback collection periods generate simultaneous access from multiple classes and departments. The system must scale to accommodate institutional growth without degradation in user experience.

- **NFR-PR-03**: At least 90% of database queries shall execute within 500 milliseconds under normal operating conditions.
    - This applies to:
        - Retrieval of subject-class pairings
        - Retrieval of feedback forms
        - Retrieval of aggregated results

    - **Rationale:**
      Fast database response ensures smooth UI rendering and prevents bottlenecks in feedback submission and dashboard loading.

- **NFR-PR-04**: The system shall complete automated sentiment analysis and generation of aggregated summaries for up to 10,000 feedback comments within 2 hours after the feedback period closes.
    - **Timing Relationship:**
      Feedback period close time → Batch processing initiation → Sentiment classification → Aggregated report generation → Faculty dashboard availability.

    - **Rationale:**
      Timely processing ensures faculty members can access analysed results within 24 hours, supporting rapid teaching improvements and administrative reporting.

5.2. Safety Requirements

The SFRS shall include safeguards to prevent institutional data loss, irreversible administrative actions and misuse that could result in operational disruption or governance failure. Although the system does not involve physical safety risks, it must ensure protection against data integrity loss and procedural harm.

- **NFR-SF-01**: The system shall perform automated daily backups of the database and support point-in-time recovery.
    - **Safeguard:**
      Backup logs shall be maintained and recovery procedures documented.

    - **Harm Prevented:**
      Prevents permanent loss of institutional data due to system failure, software error or infrastructure disruption.

Regulatory Considerations
Although no physical safety certification is required, the system shall operate in accordance with institutional governance policies and data management controls as outlined in the PS.

No external safety certifications (e.g., medical or industrial safety standards) are required for this system.

5.3. Security Requirements

The Student Feedback Review System shall implement security controls to ensure confidentiality, integrity, and regulatory compliance in the handling of student and faculty data.

- **NFR-SEC-01**: The system shall comply with Singapore's Personal Data Protection Act (PDPA) in the collection, storage, processing, and disclosure of personal data.
- **NFR-SEC-02**: All data transmitted between client devices and the application server shall be encrypted using TLS 1.3 or higher.
- **NFR-SEC-03**: Sensitive data stored in the database shall be encrypted using AES-256 encryption or equivalent.
- **NFR-SEC-04**: Administrative audit logs shall be protected against unauthorized modification or deletion.
- **NFR-SEC-05**: The database shall not be directly accessible from the public internet and shall only be accessible through authorized application services.

Security Certifications
No formal third-party security certification (e.g., ISO 27001 certification for the product itself) is mandated in the PS. However, compliance with PDPA is mandatory as stated under Section 4.4 of the PS.

5.4. Software Quality Attributes

This section defines the software quality attributes of the system, organised according to McCall's Software Quality Model. The attributes are grouped under three perspectives: Product Operation, Product Revision and Product Transition. Only the quality factors relevant to this system are selected. The link to the full **iRTM spreadsheet** can also be found here.

Clarification of Attribute Preferences
In alignment with the PS goals:

- Availability and Scalability are prioritized over advanced customization.
- Ease of use and accessibility are prioritized over feature richness.
- Data reliability and integrity are prioritized over rapid feature iteration.

5.4.1 Product Operation Perspective

These attributes describe how well the system performs in its operational environment.

5.4.1.1 Reliability

- **NFR-QUAL-01**: The system shall maintain 99.9% operational availability during active academic semesters, excluding scheduled maintenance windows.
- **NFR-QUAL-02**: Successfully submitted feedback shall remain retrievable after system restart, crash recovery, or database failover.

5.4.1.2 Efficiency

- **NFR-QUAL-03**: The system architecture shall support scaling to accommodate institutional growth up to at least 8,000 enrolled students without requiring major architectural redesign.

5.4.1.3 Usability

- **NFR-QUAL-04**: At least 90% of first-time student users shall complete feedback submission within 10 minutes without external assistance during usability testing.

5.4.2 Product Revision Perspective

These attributes describe how well the software tolerates internal changes.

5.4.2.1 Maintainability

- **NFR-QUAL-05**: The system shall maintain unit test coverage of at least 70% for core business logic modules.

5.4.3 Product Transition Perspective

These attributes describe how well the software adapts to external environments and systems.

5.4.3.1 Interoperability

- **NFR-QUAL-06**: The system shall integrate with external institutional databases (e.g., registrar and HR systems) using defined API interfaces for synchronization of student and faculty records.

5.5. Business Rules

The SFRS shall operate according to the following institutional governance principles. These business rules define operational constraints and role-based conditions under which system functions are permitted.

- **BR-01**: Only users assigned an institutional role (Student, Faculty, Administrator) shall access system functions permitted for that role.
  - **Principle:**
    System functions are governed strictly by assigned roles.

- **BR-02**: Only Academic Administrators may create, update, deactivate, or assign roles to user accounts.
  - **Principle**:
    Role governance authority resides exclusively with Academic Administrators.

- **BR-03**: At most one feedback period may be active per academic term.
  - **Principle:**
    Ensures procedural consistency and prevents overlapping evaluation cycles.

- **BR-04**: Feedback submissions are permitted only within the defined start and end dates of the active feedback period.
  - **Principle:**
    Maintains fairness and institutional standardization.

- **BR-05**: A student may submit feedback only once per assigned subject-class pairing per feedback cycle.
  - **Principle:**
    Prevents duplicate submissions and ensures data integrity.

- **BR-06**: Faculty members may only access aggregated feedback results associated with subjects and classes to which they are assigned.
  - **Principle**:
    Maintains student anonymity and preserves fairness in review processes.

- **BR-07**: Feedback questions shall not be modified or deleted once a feedback period has commenced.
  - **Principle:**
    Ensures data consistency and comparability of responses.

- **BR-08**: Only Academic Administrators may access administrative audit logs.
  - **Principle:**
    Preserves accountability and restricts sensitive governance information.

- **BR-09**: Feedback collected through the system shall be used solely for academic quality assurance and teaching improvement purposes.
    - **Principle:**
      The system shall not be used for automated grading or disciplinary action without separate institutional review.

6. Other Requirements

6.1. Data Retention and Archival

- **OR-DA-01:** Feedback data shall be retained for a minimum institutional retention period as defined by administrative policy.
- **OR-DA-02:** Past feedback records shall remain accessible for reporting purposes after feedback periods close.
- **OR-DA-03:** Archived feedback data shall not be modifiable.
- **OR-DA-04:** Deletion of student or faculty accounts shall not remove historical feedback records.

6.2. Internationalisation

- **OR-IN-01:** The system interface shall support standard English language usage.
- **OR-IN-02:** The system shall display dates and times in institutional format.

6.3. Legal

- **OR-LE-01:** Personal data shall only be accessible to authorised roles.
- **OR-LE-02**: Audit logs shall record administrative account modifications.

6.4. Reuse Objectives

- **OR-RO-01:** The system architecture shall allow future expansion to support additional feedback categories.
- **OR-RO-02:** The questionnaire module shall support addition of new question types without restructuring existing data.

6.5. Version Control

A version control method shall be used to manage changes and maintain traceability of project artefacts. All controlled project documents shall follow a structured version numbering scheme and each update shall be recorded in the Change History section.

6.5.1. Documents Version Control

- **OR-VCD-01:** Major versions (v1.0, v2.0) shall indicate significant changes to requirements or adding new features.
- **OR-VCD-02:** Minor versions (v1.1, v1.2) shall indicate slight incremental updates, refinements or requirement clarifications that do not affect the project scope.
- **OR-VCD-03:** Revision numbers (v1.1.1) shall indicate editorial changes, formatting or typographical fixes that do not affect the system behaviour.

6.5.2. Source Code Version Control

- **OR-VCC-01:** Source code versions shall follow the semantic versioning format: MAJOR.MINOR.PATCH

- **MAJOR:** Incompatible architectural or functional changes
- **MINOR:** Backward-compatible feature additions or enhancements
- **PATCH:** Defect fixes or non-breaking updates

- **OR-VCC-02:** All source code shall be stored in a version controlled repository with recorded commit history.

6.5.3. Source Code Configuration Management

- **OR-VCC-03:** The repository shall implement a controlled branching model:
  - main: Production ready and tested code.
  - develop: Integration of completed features prior to release.
  - feature/{id-feature-name}: Isolated developments of new features branches using the format ” (e.g., “feature/FR-AC-01-role-access”).
  - hotfix: correction of critical production defects.

- **OR-VCC-04:** All commits shall follow a structured commit message format to preserve traceability.
  - E.g. [FR-AC-01] Implemented role based access logic.

- **OR-VCC-05:** Commit messages shall clearly describe the scope of changes and avoid ambiguous wording.
- **OR-VCC-06:** Production releases shall be tagged using the defined semantic version number (e.g., v1.0)

6.6. Change Mitigation Controls

6.6.1. Anticipatory Contingency Requirements

The change mitigation strategies outlined in Section 2.7: Change Mitigation Strategy shall be managed in accordance with the configuration management framework described in this section.

- **OR-CM-01:** The system shall enforce the updated role definitions and access control rules while providing a controlled transition period allowing affected users to re-verify identity or update credentials without interrupting system availability.
  - **Activation Condition:** ETIS formally updates credential, authentication, or permission policies governing system access.
  - **Impact Scope:** Authentication mechanisms, role-based access control, security policy compliance.

- **OR-CM-02:** The system shall activate a temporary datastore to store and retrieve required student, faculty, and class records while logging administrative override actions to maintain audit traceability.
  - **Activation Condition:** ETIS modifies, migrates, or replaces the external database systems providing student, faculty, or class records, resulting in temporary disruption to the existing data integration interface.
  - **Impact Scope:** External system integration, data synchronisation, temporary datastore management, and audit logging.

- **OR-CM-03:** The system shall support configuration of a new institutional authentication provider while preserving existing user role associations and access permissions.
  - **Activation Condition:** ETIS replaces or upgrades its institutional identity management or authentication provider used for system login.
  - **Impact Scope:** Authentication integration, user identity mapping, role-based access control

- **OR-CM-04:** The system shall support administrative updates to course identifiers, subject-class mappings, and faculty associations without modifying previously submitted feedback records.
    - **Activation Condition:** ETIS restructures academic programmes, course codes, or subject-class mappings within the institutional academic registry.
    - **Impact Scope:** Course management integration, feedback attribution, historical data integrity.

- **OR-CM-05:** The system shall update the active feedback period end date and allow continued feedback submissions until the revised end date while ensuring previously submitted responses remain unchanged.
    - **Activation Condition:** ETIS formally extends the feedback collection period after it has already been activated.
    - **Impact Scope:** Feedback submission controls, report generation scheduling, and submission timestamp validation.

7. SRS Sign-off

This section confirms that the Software Requirements Specification for the Student Feedback Review System has been reviewed and validated by the responsible project stakeholders. The document establishes the agreed baseline of functional and non-functional requirements that will guide subsequent system design, implementation, testing, and deployment activities.

By signing below, the Project Manager confirms that:

- The Software Requirements Specification (Version 1.0) is complete and internally consistent.
- All requirements have been reviewed and validated against stakeholder expectations.
- The documented requirements establish the approved baseline for system development.
- Any subsequent changes to the requirements shall follow the Version Control procedures defined in this document.

Loy Ze Ying	Loy Ze Ying	6/3/2025
Name (Project Manager)	Signature	Date

The following authorised representatives of EduTech Institute Singapore (ETIS) confirm that they have reviewed and approved the Software Requirements Specification (SRS) Version 1.0 for the Student Feedback Review System. This document establishes the agreed requirements baseline for system development.

Name (Board of Directors Representative)	Signature	Date
Name (Chief Information Officer)	Signature	Date

8. Appendix

Appendix A: Glossary

Acronyms and Abbreviations

Acronym	Definition	Description
AA	Account Access	User authentication and credential functions allowing authorised users to access the system
AC	Access Control	Security mechanism that restricts system access and permitted actions based on authenticated user identity and assigned roles
AES	Advanced Encryption Standard	Encryption standard used to protect sensitive data such as VPNs, disk encryption and secure communications.
AL	Audit Logging	Mechanism for recording privileged administrative actions for accountability
BR	Business Rule	Institutional principle defining operational constraints under which system functions are permitted
BOR	Business Operation Requirement	High-level business requirement describing operational expectations prior to functional decomposition
DR	Data Requirement	Requirement specifying the structure, integrity, storage or retention constraints of system data
ETIS	EduTech Institution Singapore	The institutional context for which the Student Review Feedback System is developed
FA	Feedback Access	Functional capability allowing faculty members to view aggregated feedback results
FP	Feedback Period	Defined time window during which students are permitted to submit feedback
FQ	Feedback Question	Configurable qualitative or quantitative question presented to students for evaluation
FR	Functional Requirement	Specific system behaviour or capability that must be implemented and is verifiable
FS	Feedback Submission	Process by which students submit evaluation responses for assigned subject class pairings
GUI	Graphical User Interface	Visual interface that enables users to interact with the system through graphical components
IEEE	Institute of Electrical and Electronics Engineers	Professional standards organisation providing the SRS documentation template and structural guidelines

IT	Technical Requirement	Requirement defining technical specifications necessary for system operation
NFR	Non-Functional Requirement	Requirement describing system quality attributes such as performance, security or availability
OR	Other Requirement	Supplementary requirement not classified as functional or nonfunctional, such configuration controls
PDPA	Personal Data Protection Act	Singapore legislation regulating the collection, use and protection of personal data
PR	Performance Requirement	Requirement specifying measurable system performance thresholds such as response time or concurrency limits
QUAL	Quality	System characteristic relating to maintainability, reliability, scalability, usability or robustness
RBAC	Role-Based Access Control	Security model restricting system access based on assigned user roles
SEC	Security	Requirement defining confidentiality, integrity, authentication, and access protection controls
SA	System Accessibility	Requirement ensuring that the system can be accessed across supported platforms, devices and browsers while maintaining consistent functionality
SI	System Integration	Requirement describing the system ability to interface and exchange data with external institutional systems
SR	Safety Requirement	Requirement ensuring protection against data loss, misuse or operational harm within the system
SFRS	Student Review Feedback System	The institutional academic feedback management system specified in this SRS
TLS	Transport Layer Security	Cryptographic protocol used to secure data transmitted between client and server
ITR	Technical Requirement	Requirement describing infrastructure, hosting, deployment or operational constraints
UAT	User Acceptance Testing	Stakeholder validation process confirming system readiness prior to deployment
UC	Use Case	Description of an interaction between a user and the system that achieves a specific goal or system function
UR	User Requirement	Requirement describing system capabilities or constraints from the perspective of end users, focusing on usability and interaction behaviour

Defined Terms

Term	Description
Academic Administrator	Institutional staff member responsible for configuring feedback periods, managing user accounts and overseeing administrative operations
Academic Term	Defined instructional period within the academic calendar during which subjects are delivered and feedback may be collected
Authorised User	Any ETIS member (student, faculty, administrator) who is provisioned access based on institutional role and records.
Subject-Class Pairing	Association between a subject, a specific class and the assigned faculty member responsible for teaching that class
Aggregated Feedback Report	Anonymised summary of feedback results generated after the feedback period closes for faculty review
Faculty Member	Teaching staff responsible for delivering an academic subject and eligible to receive aggregated feedback reports
Institutional Content Validation Rules	Institutional guidelines used to detect and prevent inappropriate or non-compliant feedback submissions
Sentiment Analysis	Automated classification of qualitative feedback comments to identify positive, neutral or negative trends

Appendix B: System Features Response Sequences

Role-Based Access Control Response Sequence

User Request/Action	System Response
Administrator provisions a new user account with a specified role	System activates the account and applies the role's access restrictions immediately
User attempts to access a system function	System checks the user's assigned role; if permitted, access is granted; if not permitted, access is denied and an appropriate restriction message is displayed
Administrator updates an existing user's role	System applies the updated role and revises access restrictions immediately
Administrator deactivates a user account	System disables the account and revokes all associated access privileges
Administrator attempts to assign more than one role to a user	System rejects the assignment and returns a single-role constraint error

Table 1: Role-Based Access Control Response Sequence

User Account Access Response Sequence

User Request/Action	System Response
User navigates to self-service account settings and requests a credential change	System presents an identity verification prompt
User provides valid identity verification	System presents the credential update form
User submits new credentials that meet institutional complexity and expiry requirements	System confirms the update and applies the new credentials
User submits credentials that fail policy requirements (e.g., insufficient complexity, reuse of recent password)	System rejects the update and returns a specific error message describing the policy violation
User attempts to modify their own role or access privilege through any self-service function	System denies the action and returns an access restriction message

Table 2: User Account Access Response Sequence

Administrative Audit Logging Response Sequence

User Request/Action	System Response
Administrator performs any action on a user account (e.g., creates, updates, deactivates)	System automatically generates an audit record capturing the administrator identity, timestamp, and action description without requiring manual input
Administrator requests to view audit logs	System authenticates the requester's Administrator role and presents the relevant audit records
Non-administrator user attempts to access audit logs	System denies access and returns a permission error message

Table 3: Administrative Audit Logging Response Sequence

Feedback Period Management Response Sequence

User Request/Action	System Response
Administrator creates a feedback period by specifying a start date and end date for an academic term	System validates that no other period is active for the same term; if valid, the period is created and applied uniformly across all subjects and classes; if a conflict exists, the system rejects the request with a conflict error
System reaches the defined end date of an active feedback period	System automatically closes the period and permanently disables all associated feedback submission access
Administrator attempts to create a second active feedback period for the same term while one is already active	System rejects the request and returns a single-active-period constraint error

Table 4: Feedback Period Management Response Sequence

Feedback Question Management Response Sequence

User Request/Action	System Response
Administrator defines feedback questions prior to activating the feedback period	System saves the questions and associates them with the pending feedback period
Administrator marks a question as mandatory	System records the mandatory designation and applies it as a submission completion requirement
Administrator attempts to modify or delete a question after the feedback period has been activated	System denies the action and returns a data integrity restriction message specifying that questions are locked once a period is active
Administrator attempts to submit a feedback period configuration with no questions defined	System rejects activation and returns a validation error requiring at least one question

Table 5: Feedback Question Management Response Sequence

Feedback Submission by Students Response Sequence

User Request/Action	System Response
Student accesses the feedback submission section during an active feedback period	System presents only the subject-class pairings to which the student is enrolled and for which feedback has not yet been submitted in the current period
Student selects a subject-class pairing, completes all feedback questions, and submits	System validates mandatory question completion and content; if all checks pass, submission is accepted and a confirmation is displayed
Student submits without completing one or more mandatory questions	System rejects the submission and highlights the incomplete mandatory fields
Student's submission fails content validation (e.g., prohibited content detected)	System rejects the submission and returns a content validation error message without specifying which rule was violated
Student attempts to submit feedback when no feedback period is active or the period has closed	System denies access to the submission form and displays a period-status message
Student attempts to submit feedback for a subject-class pairing for which they have already submitted in the current period	System denies the action and notifies the student that a submission for that pairing already exists

Table 6: Feedback Submission by Students Response Sequence

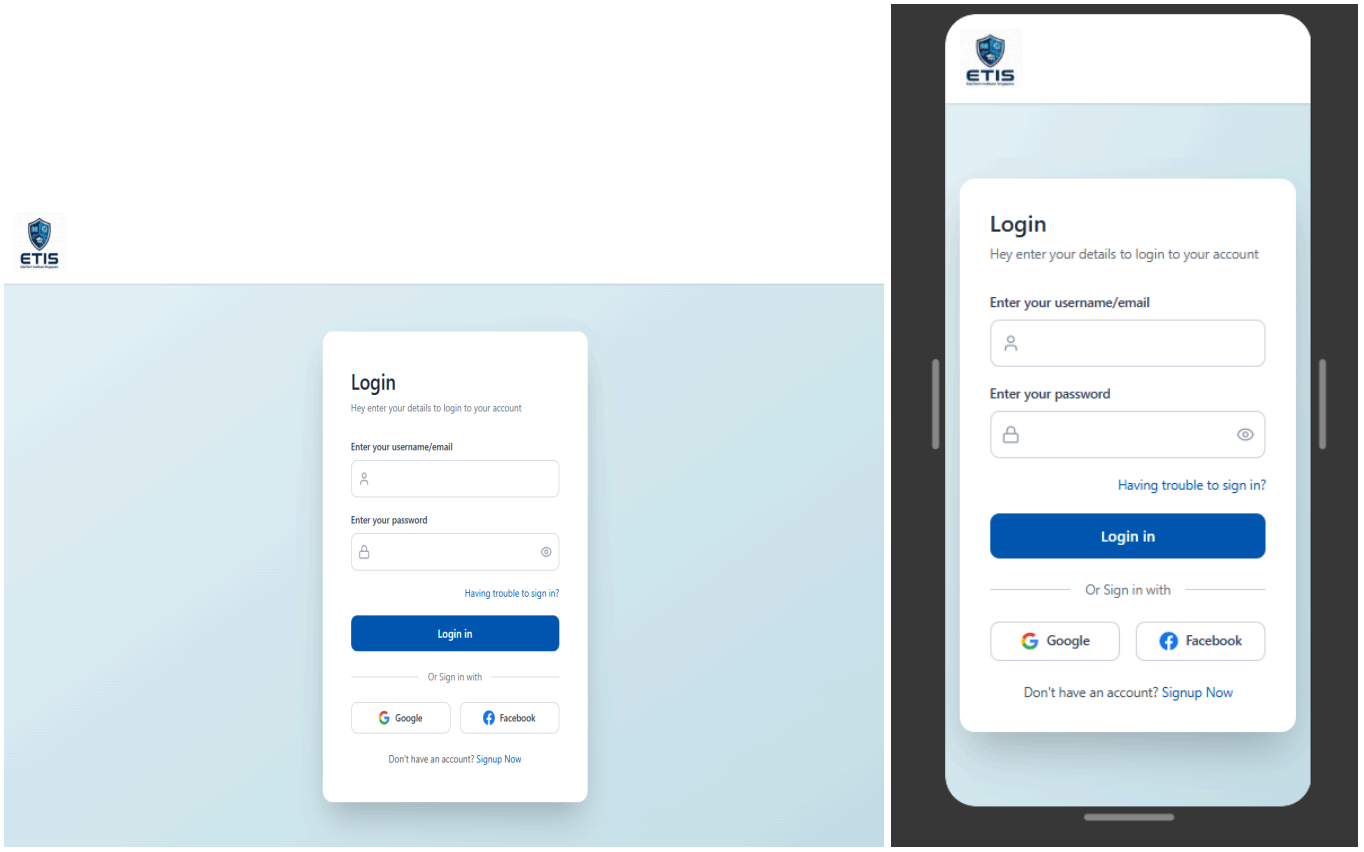
Faculty Feedback Access Response Sequence

User Request/Action	System Response
Faculty member accesses the feedback results section	System checks whether the feedback period is closed. If closed, results for all assigned subject-class pairings are presented. If the period is still active, access is denied with a period-status message
Faculty member views their feedback results	System presents aggregated data without revealing any individual student identity
Faculty member views the analysed summary	System presents the pre-generated summary for the faculty member's assigned subject-class pairings
Faculty member attempts to modify or delete any feedback record	System denies the action and returns a permission restriction message
Faculty member attempts to access feedback results for a subject-class pairing not assigned to them	System denies access and returns a scope restriction message

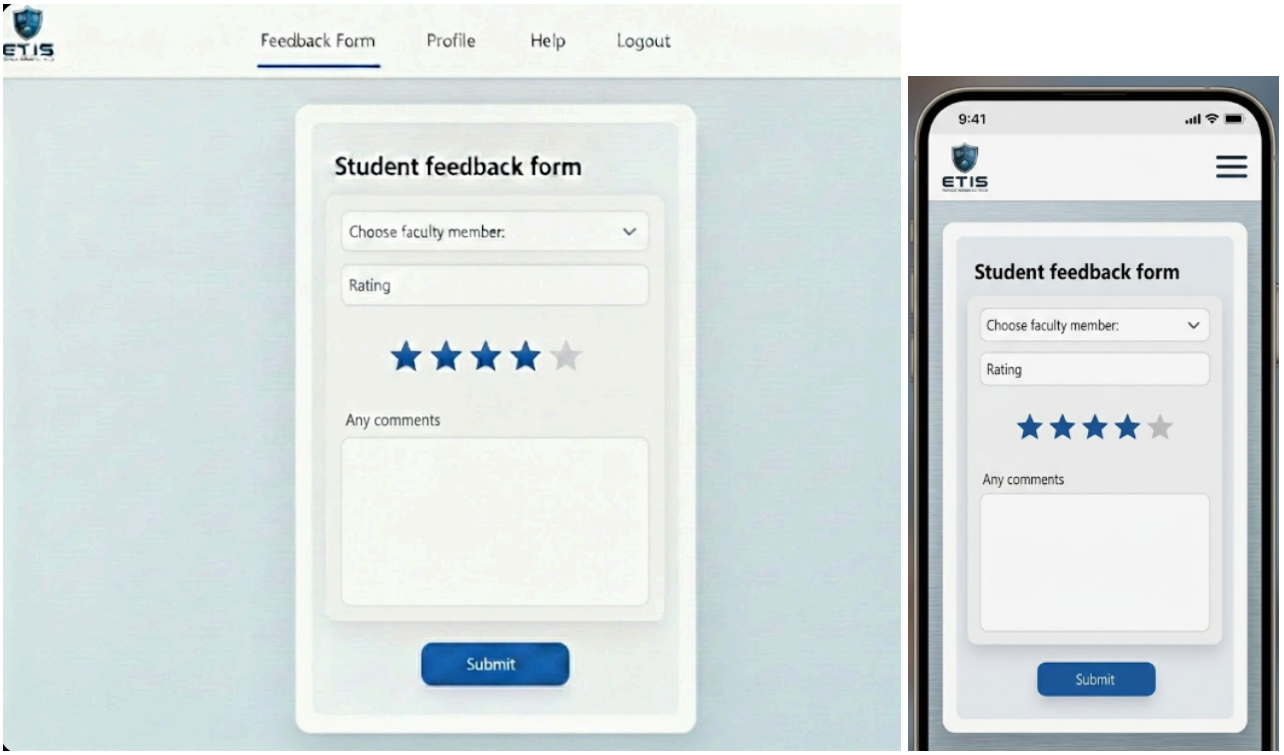
Table 7: Faculty Feedback Access Response Sequence

Appendix C: Sample Screen Image

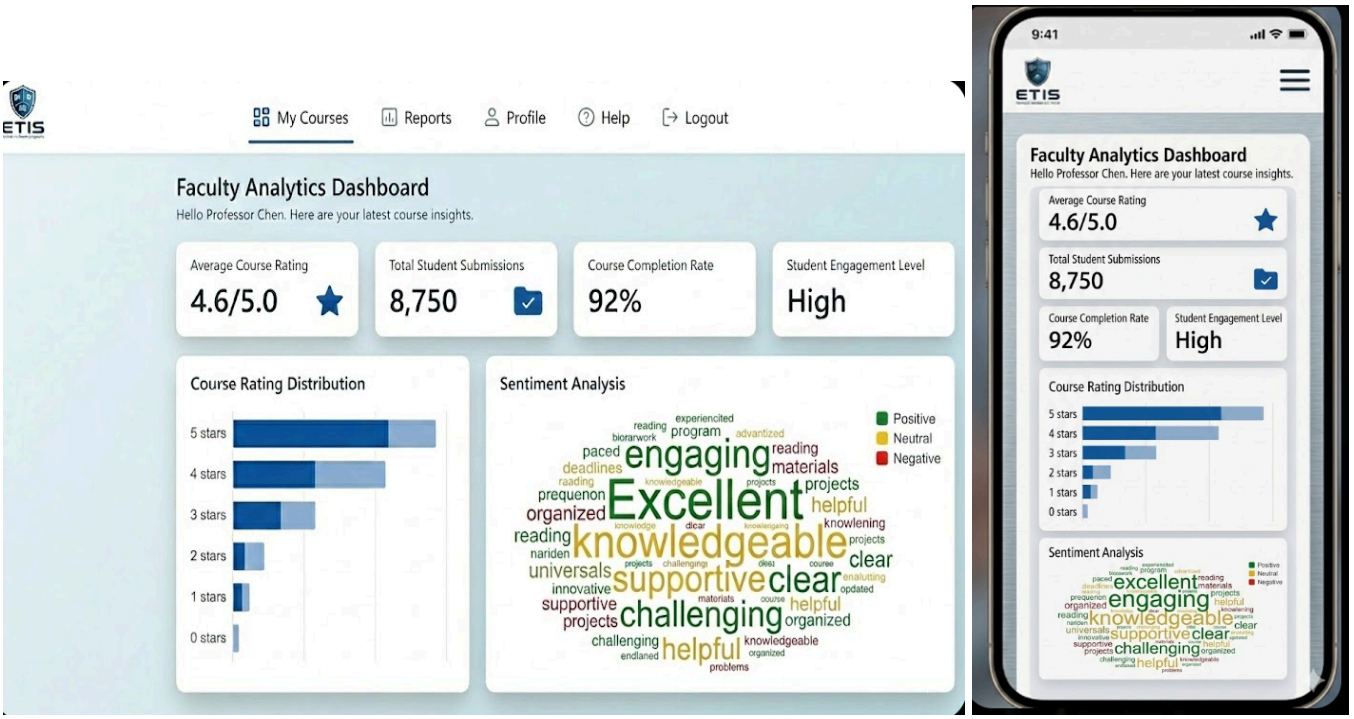
Login page for both mobile and desktop



Student feedback form page for both mobile and desktop



Faculty dashboard for both mobile and desktop



Admin dashboard for both mobile and desktop

Appendix D: Declarations

AI Usage Declaration

This document was produced by the author(s) and reviewed by the project team. Generative AI tools (e.g., ChatGPT) were used as a writing support aid to improve clarity, structure, grammar, and consistency with the IEEE 830 / Wiegers-style SRS format. AI assistance was used to:

- refine wording into clear, atomic, testable “shall” statements,
- improve consistency of requirement IDs and section phrasing,
- suggest neutral phrasing for non-functional requirements and business rules,
- check readability and reduce ambiguity.

AI tools were **not** used to fabricate stakeholder decisions, invent institutional policies, or replace requirements elicitation and analysis. All final requirements, priorities, and referenced constraints were validated by the author(s) against the project specification, analysis artefacts, and team decisions.

Examples of prompts used (representative):

- “Rewrite these requirements as atomic and testable ‘shall’ statements, keeping the original meaning and role constraints.”
- “Check this SRS section for ambiguity and rewrite to be measurable (response time, uptime, concurrency), without changing scope.”
- “Make the wording consistent with IEEE 830 conventions and keep the tone formal and objective.”
- “Suggest a clearer structure for the System Features section with description, priority, response sequences, then functional requirements.”
- “Help standardize requirement IDs (FR-XX / NFR-XX / OR-XX) and ensure naming is consistent across sections.”
- “Shorten this paragraph while keeping key constraints and references intact.”

Plagiarism Declaration

The author(s) declare that this Software Requirements Specification is an original work produced for the Student Feedback System project. Except where properly referenced, the content in this document is written in the author(s)' own words and reflects the project team's analysis and interpretation of stakeholder needs and constraints.

Where external standards, laws, or references informed the document (e.g., IEEE Std 830-1998, Karl E. Wiegers' SRS guidance, and Singapore PDPA), they are acknowledged in the **References** section, and any reused ideas are adapted for this project context. No uncredited copying from other student submissions, online templates, vendor documents, or published specifications has been included.

All team members contributed to review and validation of the requirements content, and the final version represents the agreed baseline for development under the project's configuration and change control process.

Appendix E: Change Control Agreement

CHANGE CONTROL AGREEMENT

Project: Student Feedback Review System (SFRS)
Vendor: NovaStack Pte Ltd
Client: EduTech Institute Singapore (ETIS)
Effective Date: [Insert Date]

1. Purpose

This Change Control Agreement (CCA) defines the formal process by which changes to the approved Software Requirements Specification (SRS) baseline shall be requested, evaluated, approved, rejected, or deferred during the Software Development Life Cycle (SDLC).

This agreement ensures scope stability, traceability, cost control, and schedule governance.

2. Scope of Change Control

This agreement applies to:

- Modifications to approved functional or non-functional requirements
- Introduction of new requirements not traceable to the SRS baseline
- Changes affecting system architecture, integrations, or data structures
- Changes impacting cost, timeline, compliance, or project deliverables

Minor editorial corrections that do not alter requirement meaning do not require formal change control.

3. Change Request Submission

All proposed changes shall be submitted through a **formal Change Request (CR)** document containing:

- Description of proposed change
- Business justification
- Affected requirement(s), if applicable
- Requested priority
- Requested timeline

Change Requests must be submitted in writing by an authorised representative of ETIS or NovaStack.

4. Impact Analysis

Upon receipt of a Change Request, NovaStack shall conduct a Change Impact Analysis covering:

- Technical feasibility
- Architectural impact
- Impact on existing requirements
- Cost implications
- Schedule implications
- Risk assessment
- Quality and compliance considerations

Findings shall be documented in a **Change Impact Report (CIR)**.

5. Classification of Changes

Change Requests shall be classified as:

- **Modification** - Adjustment to existing approved requirement
- **New Requirement** - Functionality not derivable from current SRS baseline
- **Deferred Enhancement** - Accepted but postponed to future release
- **Rejected Change** - Not approved for implementation

6. Approval Process

A Change Request shall only proceed upon:

- Review of the Change Impact Report
- Agreement by the Project Manager (NovaStack)
- Written approval from the ETIS authorised representative

If the change:

- Falls outside original contractual scope, or
- Requires additional resources beyond agreed allocation

It shall be treated as a **contract variation** and may require adjustment to cost and/or schedule.

No development work shall commence without formal written approval.

7. Baseline and Documentation Updates

Upon approval:

- The SRS version number shall be incremented
- The Requirements Traceability Matrix (iRTM) shall be updated
- A unique Requirement ID shall be assigned if applicable
- Affected design, implementation, and test artefacts shall be revised

All changes shall be logged in the Project Version History.

8. Governance and Authority

Only designated representatives listed below are authorised to approve changes:

NovaStack Representative:
Name: _____
Title: _____
Signature: _____

ETIS Representative:
Name: _____
Title: _____
Signature: _____

9. Effective Control

This Change Control Agreement remains in effect throughout the project lifecycle and governs all requirement modifications after SRS baseline approval.

Annex A: Data Dictionary

(Provided by ETIS in Project Specification)

A. Core Entities

1) UserAccount

Attribute	Example	Description
username	‘John Doe’	Login username using institutional credentials.
password_hash	‘b94d27b9934d3e08a52e52d7da7dabfac484efe37a5380ee9088f7ace2efcde9’	Encrypted password for authentication.
role	‘Student’	User role determining system access privileges.

B. Feedback Collection Entities

1. Question

Attribute	Example	Description
question_text	‘On a scale of 1-5 what is your rating on this class?’	Text of the feedback question.
question_type	‘Rating’	Type of response expected.
is_required	True	Indicates whether the question is mandatory.

2. FeedbackSubmission

Attribute	Example	Description
feedback_id	1	Unique identifier for a feedback submission.
student_id	200	Student who submitted the feedback.
faculty_id	354	Faculty being reviewed.
submitted_at	'2024-01-29 10:08:00'	Timestamp of submission.

C. Analytics / Reporting Entities

1. FeedbackAnalysisSummary

Attribute	Example	Description
sentiment_result	'Positive', 'Neutral', 'Negative'	Overall sentiment classification.
aggregated_summary	‘Overall, the feedback received has been…’	Aggregated summary on all the feedback received by students, generated by ML.

Figure 1: Context Diagram

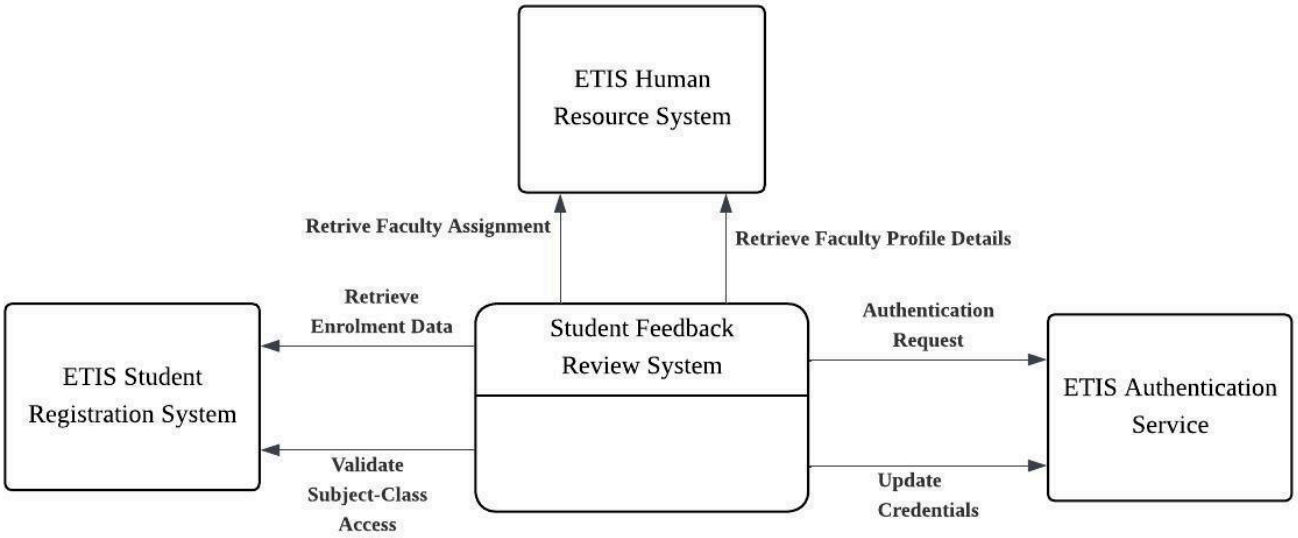


Fig 1: Context Diagram

Figure 2: High Level Data Flow

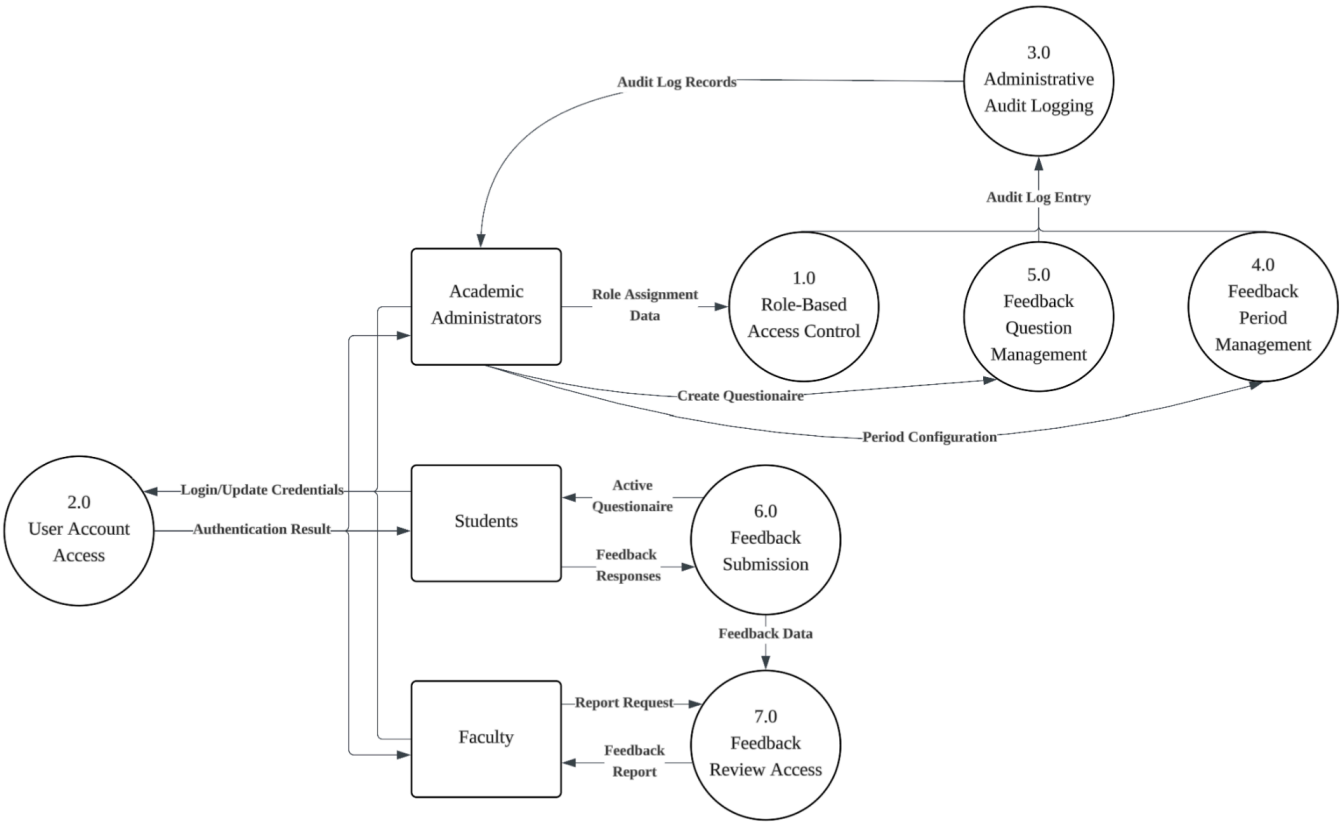


Fig 2: High Level Data Flow

Figure 3: Use Case Diagram

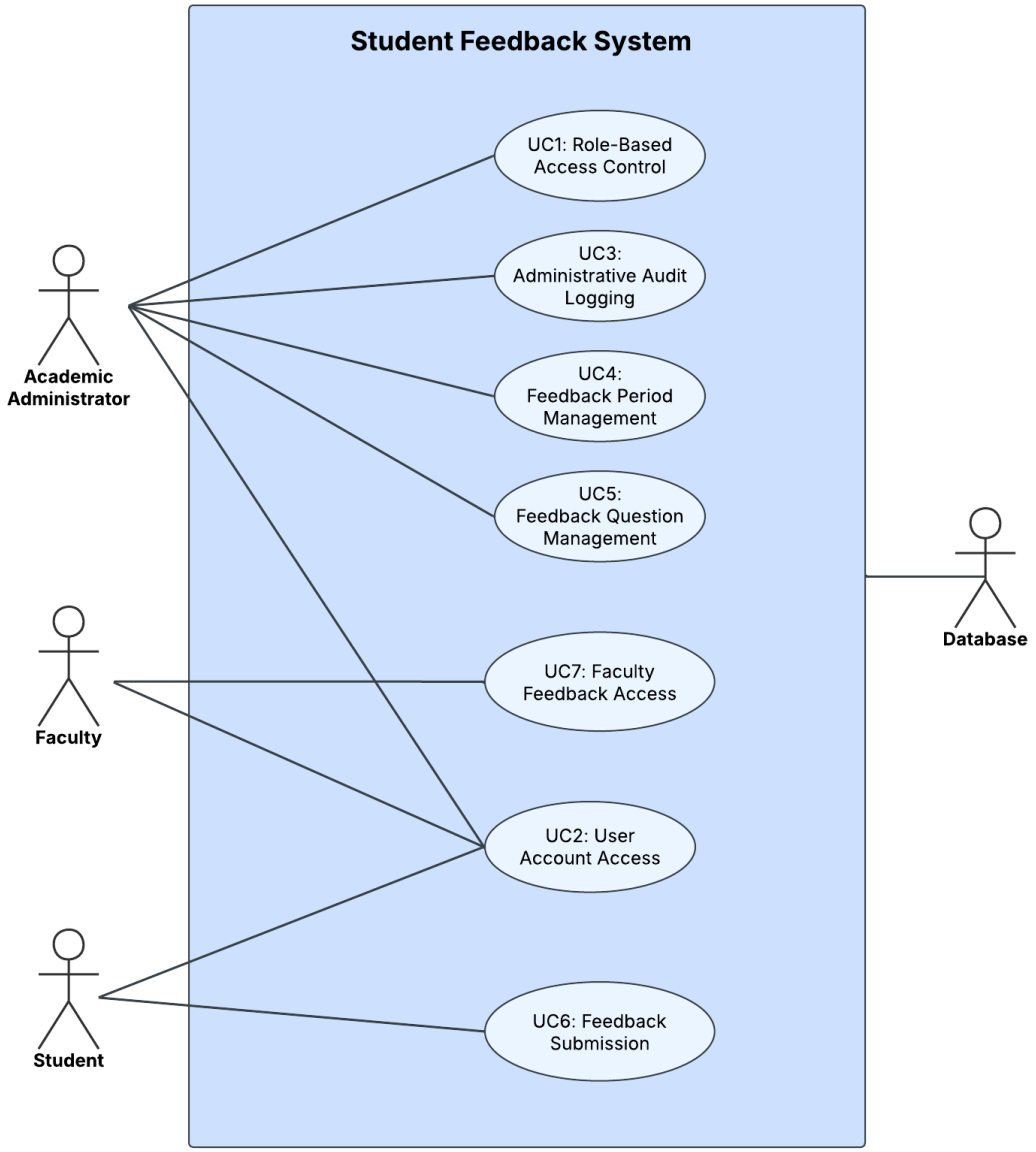


Fig 3: Use Case Diagram (Revised from Project Specifications)